

1 The Multidimensional Case

2 Now we consider the case where $\theta = (\theta_1, \theta_2, \dots, \theta_p)$.

3 We define the **Fisher Information matrix** to be the p by p matrix whose (i, j)
4 entry is

$$5 \quad I_{ij}(\theta) = E \left[-\frac{\partial^2}{\partial \theta_i \partial \theta_j} \log f(x; \theta) \right]$$

6 **The Main Result:** Under appropriate "regularity conditions," if $X_1, X_2, \dots, X_n$
7 are i.i.d. $f(x; \theta)$ then

$$8 \quad \sqrt{n}(\hat{\theta} - \theta_0) \longrightarrow N_p(0, \mathbf{I}^{-1}(\theta_0))$$

9 $\hat{\theta}$ is the MLE for θ
Recall that N_p denotes the p -dimensional multivariate normal distribution.

10 **Practical Version:** Under "regularity conditions," $\hat{\theta}$ is approximately nor-
11 mal with variance $\mathbf{I}^{-1}(\hat{\theta})/n$ compare with $\frac{1}{n}\mathbf{I}(\hat{\theta})$

1 **Exercise:** Suppose that $X_1, X_2, \dots, X_n$ are i.i.d. $N(\mu, \sigma^2)$. Derive the joint
2 distribution of the MLE for $\theta = (\mu, \sigma^2)$.

$$3 \quad f(x; \theta) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \quad -\infty < x < \infty$$

$$6 \quad \frac{\partial^2}{\partial \mu^2} \log f(x; \theta) = -\frac{1}{\sigma^2}$$

$$9 \quad \frac{\partial^2}{\partial (\sigma^2)^2} \log f(x; \theta) = \frac{1}{2\sigma^4} - \frac{(x-\mu)^2}{\sigma^6}$$

$$12 \quad \frac{\partial^2}{\partial \mu \partial (\sigma^2)} \log f(x; \theta) = -\frac{(x-\mu)}{2\sigma^4}$$

So,

$$\begin{aligned} \underline{I}(\underline{\theta}) &= E \begin{bmatrix} \frac{1}{\sigma^2} & \frac{(X-\mu)}{2\sigma^4} \\ \frac{(X-\mu)}{2\sigma^4} & \frac{(X-\mu)^2}{\sigma^6} - \frac{1}{2\sigma^4} \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{\sigma^2} & 0 \\ 0 & \frac{1}{2\sigma^4} \end{bmatrix}. \end{aligned}$$

The MLE for $\underline{\theta}$ is $\hat{\underline{\theta}} = \begin{bmatrix} \bar{x} \\ (\frac{n-1}{n})s^2 \end{bmatrix}$.

It is approximately multivariate normal with covariance matrix of

$$\underline{I}^{-1}(\underline{\theta}_0)/n = \begin{bmatrix} \sigma_0^2/n & 0 \\ 0 & 2\sigma_0^4/n \end{bmatrix}$$

In practice, replace σ_0^2 with $\hat{\sigma}^2$

$$V(\hat{\bar{x}}) = \hat{\sigma}^2/n, \quad V\left(\left(\frac{n-1}{n}\right)s^2\right) = \frac{2\hat{\sigma}^4}{n}$$

Also, $\text{Cov}(\hat{\bar{x}}, (\frac{n-1}{n})s^2) = 0$, i.e. $\hat{\bar{x}}, (\frac{n-1}{n})s^2$ are independent

1 Example of a Numerical Approach

2 In a case where the MLE is found numerically, we can, use the ob-
 3 served Fisher Information in place of $I^{-1}(\hat{\theta})/n$ just as we did in the
 4 one-dimensional case.

5 In a multidimensional optimization using `optim()`, the procedure **mini-**
 6 **mizes** the negative of the log likelihood. Hence, the Hessian is the matrix
 7 of second derivatives evaluated at the peak, i.e.,

$$8 \quad \frac{\partial^2}{\partial \theta^2} (-\log L(\theta)) \Big|_{\theta=\hat{\theta}}$$

9 We do not need to scale this by n , and we do not need to multiply by -1 .

10 To get the approximate covariance matrix, simply invert this Hessian.

1 **Exercise:** Revisit the problem where we were numerically finding the MLE
 2 of $\theta = (\alpha, \beta)$ in the case of an i.i.d. sample from the Gamma distribution.
 3 Simulate some data, and construct the MLE and approximate the covari-
 4 ance matrix. Interpret the result.

5 In example, $n=100$, $\alpha_0=2$, $\beta_0=4$
 6 $\hat{\alpha}=1.81$, $\hat{\beta}=3.38$

7 The approx. covariance matrix for $(\hat{\alpha}, \hat{\beta})$ is
 8 solve(holout\$hessian)

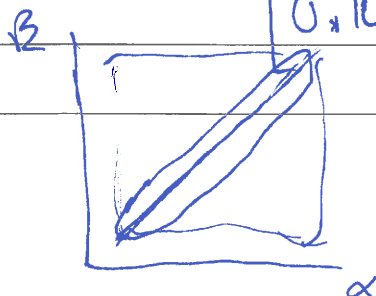
$$9 \quad \begin{bmatrix} 0.0557 & 0.104 \\ 0.104 & 0.258 \end{bmatrix} \quad SE(\hat{\alpha}) = \sqrt{0.0557} = 0.24$$

$$10 \quad SE(\hat{\beta}) = \sqrt{0.258} = 0.508$$

$$11 \quad Cov(\hat{\alpha}, \hat{\beta}) = 0.104$$

$$12 \quad Corr(\hat{\alpha}, \hat{\beta}) = 0.87$$

$$g(\alpha, \beta) = \alpha/\beta = E(X)$$



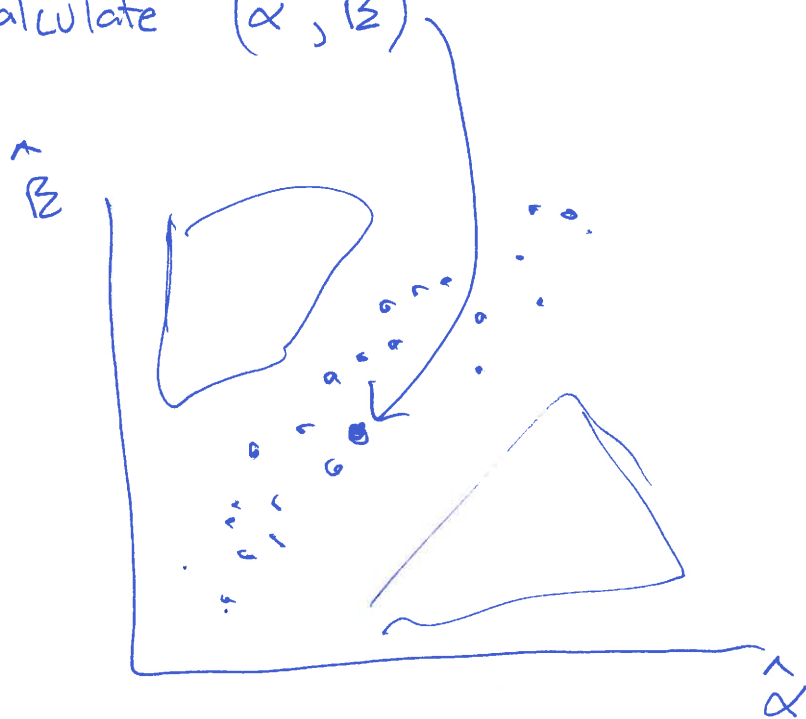
How could I arrive at an estimate of $\text{Cov}(\hat{\alpha}, \hat{\beta})$ (and hence $\text{Corr}(\hat{\alpha}, \hat{\beta})$) using only simulations?

Stick with $n=100$, $\alpha_0=2$, $\beta_0=4$

repeat $\uparrow$ many times:

sample 100 values from $\text{Gamma}(2, 4)$

calculate $(\hat{\alpha}, \hat{\beta})$



Calculate sample covariance between

$(\hat{\alpha}_1, \hat{\beta}_1), (\hat{\alpha}_2, \hat{\beta}_2), \dots, (\hat{\alpha}_r, \hat{\beta}_r)$

Should be close to 0.104

1 Multivariate Delta Method

2 We can extend the Delta method to cases where the parameter is multidimensional.
3

4 In general, suppose that $\mathbf{Y}_n$ is approximately $N_p(\boldsymbol{\mu}, \Sigma/n)$. Now consider
5 an m -dimensional function $\mathbf{g}()$ where $m \leq p$:

$$6 \quad \mathbf{g}(\mathbf{Y}_n) = \begin{bmatrix} g_1(\mathbf{Y}_n) \\ g_2(\mathbf{Y}_n) \\ \vdots \\ g_m(\mathbf{Y}_n) \end{bmatrix}$$

m-dimensional

7 Then, $\mathbf{g}(\mathbf{Y}_n)$ is approximately $N_m(\mathbf{g}(\boldsymbol{\mu}), \mathbf{G}\Sigma\mathbf{G}^T/n)$ where the (i, j) entry of
8 $\mathbf{G}$ is $\partial g_i(\mathbf{y})/\partial y_j$. This all assumes that $\mathbf{G}$ is of full rank.

1 **Exercise:** Place our parameter estimation problem into the framework of
2 the above statement of the Delta method.

3 $\mathbf{Y}_n$ is replaced with $\hat{\boldsymbol{\theta}}_{MLE}$

4 Σ is replaced with $\mathbf{I}^{-1}(\boldsymbol{\theta}_0)$

5 $\boldsymbol{\mu}$ is replaced with $\boldsymbol{\theta}_0$

6 Delta method says that $\mathbf{g}(\hat{\boldsymbol{\theta}})$ is approx.
7 normal with mean $\mathbf{g}(\boldsymbol{\theta}_0)$ and covariance
8

9 $\mathbf{G}\mathbf{I}^{-1}(\boldsymbol{\theta}_0)\mathbf{G}^T/n$, $\mathbf{G} = \begin{bmatrix} \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_1} & \frac{\partial g_1(\boldsymbol{\theta})}{\partial \theta_2} & \dots \\ \frac{\partial g_2(\boldsymbol{\theta})}{\partial \theta_1} & \dots & \dots \\ \vdots & \ddots & \ddots \end{bmatrix}$

Example

$X_1, X_2, \dots, X_n$ iid $\text{Gamma}(\alpha, \beta)$

We know how to find the MLE for $\underline{\theta} = (\alpha, \beta)$,
and also how to find its Fisher Info. matrix
(using Hessian)

Suppose want to estimate $E(X_i) = \frac{\alpha}{\beta}$

Invariance prop. says MLE for $E(X_i)$ is $\frac{\hat{\alpha}}{\hat{\beta}}$

But: How to attach a SE to this estimator?

Use multivariate Delta method.

Note $m=1$, i.e. $g_1(\underline{\theta}) = \frac{\alpha}{\beta}$

$\underline{G}$ is 1 by 2

$$\underline{G} = \begin{bmatrix} \frac{\partial g_1(\underline{\theta})}{\partial \alpha} & \frac{\partial g_1(\underline{\theta})}{\partial \beta} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{\beta} & -\frac{\alpha}{\beta^2} \end{bmatrix}$$

The Delta method says that $g_1(\hat{\theta}) = \hat{\alpha}/\hat{\beta}$ is approx. normal with covariance "matrix" is

$$\underline{\underline{G}} \underline{\underline{I}}^{-1}(\underline{\underline{\theta}}_0) \underline{\underline{G}}^T / n = \underline{\underline{G}} \underbrace{(\underline{\underline{I}}^{-1}(\underline{\underline{\theta}}_0) / n)}_{\text{approx. using the inverse of Hessian}} \underline{\underline{G}}^T$$

approx. using the ~~Hessian~~
inverse of Hessian

So, $\underline{\underline{I}}^{-1}(\underline{\underline{\theta}}_0) / n$ is approx. using software

$$\underline{\underline{G}} \text{ is approx. using } \begin{bmatrix} 1/\hat{\beta} & -\hat{\alpha}/\hat{\beta}^2 \end{bmatrix}$$

Hence, the approx. covar. "matrix" is

$$\begin{bmatrix} 1/\hat{\beta} & -\frac{\hat{\alpha}}{\hat{\beta}^2} \end{bmatrix} \begin{bmatrix} \text{inverse of} \\ \text{Hessian returned} \\ \text{by software} \end{bmatrix} \begin{bmatrix} 1/\hat{\beta} \\ -\frac{\hat{\alpha}}{\hat{\beta}^2} \end{bmatrix}$$

Note that it is a scalar since only estimating single parameter $E(X_i) = \alpha/\beta$